DockRush: Deadline-Aware Warehouse Fulfilment with Reinforcement Learning

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Abstract

DockRush formulates warehouse order fulfilment as a practical reinforcement-learning problem. A robot receives stochastic orders, picks items, delivers them to docks before truck departures, and manages capacity and battery constraints. We model this as an event-driven semi-Markov decision process (SMDP), where actions have different durations. The main agent is a masked PPO policy with a permutation-invariant attention encoder and pointer-style order selection. On 100 held-out seeds, PPO-Attention obtains 153.0 ± 3.9 task return, 67.3% revenue captured, and 78.3% express-order on-time delivery, outperforming DQN and evaluation baselines.

1 Problem and Unique Ideation

The practical problem is deciding which warehouse task a robot should perform next when orders arrive over time and trucks leave at uncertain deadlines. The project is intentionally RL-only: PPO, DQN, and tabular Q-learning are the learning methods; FIFO, EDF, Greedy, and Random are used only as evaluation reference policies, not as the proposed solution.

The distinctive design is the combination of five elements in one RL environment:

1. stochastic non-homogeneous order arrivals with midday demand peaks and rush shocks;
2. hidden truck-departure jitter, express/standard/bulk priorities, and order deferral;
3. variable-duration macro-actions, making the problem an SMDP rather than a fixed-step toy grid;
4. a hard action mask for invalid picks, unsafe battery moves, and empty deliveries; and
5. a permutation-invariant attention policy that can run with larger order queues than seen during training, plus a shared asynchronous multi-robot extension.

This is a system-level ideation contribution rather than a claim of a new PPO theorem: the originality is in the realistic RL formulation, the deadline-sensitive reward design, and the stress tests that measure generalisation, ablations, and fleet scaling.

2 RL Formulation

2.1 State space $\mathcal{S}$

At each decision event, the observation is a dictionary:

$$s_t = (g_t, O_t, M_t[, R_t]).$$

For the nominal single-robot setting, $g_t \in \mathbb{R}^{16}$ contains normalized time (including sine/cosine phase), battery, load, position, carried orders per dock, dock countdowns, backlog, recent arrival rate, and capacity. The order matrix is $O_t \in \mathbb{R}^{16\times 14}$: each of 16 slots contains presence, value, tier, destination dock, deadline slack after an estimated completion time, remaining items, carried/loaded items, distance, capacity feasibility, and age features. M_t is a binary action mask. In multi-robot experiments, $R_t \in \mathbb{R}^{3\times 12}$ contains robot identity, position, target, load, battery, and busy-time tokens. All continuous features are normalized and padded slots are zero.

2.2 Action space $\mathcal{A}$

The action space is discrete:

$$\begin{aligned}\mathcal{A} = &\{\text{PICK}(i) : i = 1, \dots, N_{\max}\} \\ &\cup \{\text{DELIVER}(d) : d = 1, \dots, D\} \\ &\cup \{\text{CHARGE}, \text{WAIT}\}.\end{aligned}$$

With $N_{\max} = 16$ and $D = 3$, there are 21 actions. A selected action is executed until the corresponding event completes, so its duration τ_t varies with travel distance, item count, picking, loading, charging, or waiting. The mask removes impossible or unsafe choices; masked logits are excluded before sampling.

2.3 Reward function

For one option, the environment logs fulfilled, lost, and deferred orders. The reward is

$$r_t = F_t - \lambda_m L_t - \frac{\lambda_d}{2} D_t - \lambda_c \tau_t K - \lambda_b I_t + \alpha\big(\Phi(s_{t+1}) - \Phi(s_t)\big),$$

where F_t is fulfilled order value, L_t is missed-order loss, D_t is deferred value, K is the number of robots, and I_t indicates a stranded battery. The default coefficients are $\lambda_m = 1$, $\lambda_d = 0.3$, $\lambda_c = 0.01$, $\lambda_b = 50$, and $\alpha = 0.3$. The potential Φ rewards progress toward loading valuable orders and carrying orders that can still reach their dock before departure. This encourages useful progress without replacing the task return used for final comparison.

2.4 SMDP transition and objective

The simulator uses a 720-tick shift on a grid warehouse. Orders follow a thinning-based non-homogeneous Poisson process; trucks depart every 120–180 ticks with Gaussian timing jitter. The robot has capacity 6 and battery capacity 400. The SMDP-correct choice is duration-aware discounting in GAE:

$$\gamma_t = \gamma^{\tau_t}, \qquad \delta_t = r_t + \gamma_t V(s_{t+1}) - V(s_t).$$

With $\gamma = 0.995$ per tick this spans only about 20 decisions, so the final PPO agent uses a per-decision discount $\gamma_t = \gamma$, which trained better (Section 4).

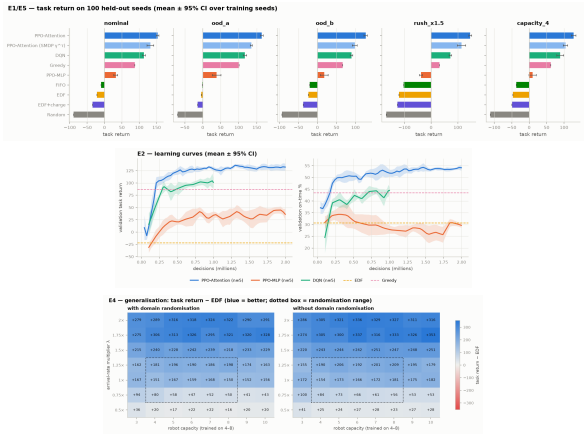


Figure 1: Benchmark, learning curves, and out-of-distribution generalisation.

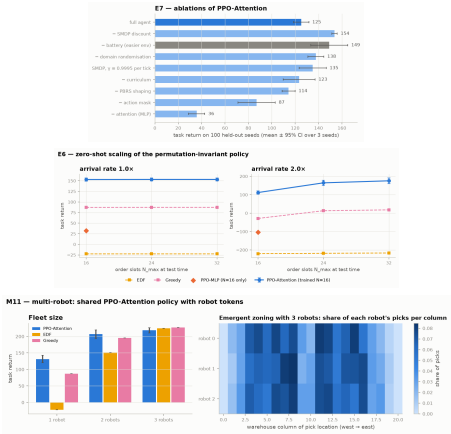


Figure 2: Ablations, zero-shot scaling, and multi-robot results.

3 Algorithms and Evaluation

The primary learner is PPO with clipped policy updates, value loss, entropy regularization, gradient clipping, vectorized environments, masked actions, and the attention encoder. The encoder treats order slots as a set and uses a learned `[CLS]` token to produce the policy context; a pointer head scores each order slot. DQN uses a dueling double-Q network, prioritized replay, n-step returns, the same observation/action mask, and the duration-aware discount. Tabular Q-learning is used only in a tiny exact-check environment.

Training uses five independent seeds for the main agents and three for ablations. Final results use 100 held-out evaluation seeds disjoint from checkpoint selection. We compare nominal performance, unseen layouts, arrival-rate/capacity grids, larger queues, six ablations, attention behavior, a tiny near-optimality check, and one-to-three robots.

4 Results and Discussion

PPO-Attention is the strongest nominal policy and retains its advantage on unseen layouts, arrival-rate changes, and capacity changes. Its express on-time rate is 78.3%, compared with 54.6% for Greedy. The attention policy also transfers from 16 training slots to 24 and 32 test-time slots. The ablation study (on the original agent) shows that removing attention reduces task return by about 89 points and removing the action mask by about 38 points, supporting the importance of set-based order selection and feasibility-aware decisions.

A per-decision discount raised task return from 132.0 ± 9.5 to 153.0 ± 3.9 and revenue from 60.2% to 67.3%; a $5\times$ larger network did not help. Look-ahead over 32 sampled futures lifts the best checkpoint from 66.4% to 71.0% revenue ($\approx$3 s/decision).

The project also tests an applied multi-robot extension with a shared policy and order claiming, trained with the original discounting. Two robots reach 207.1 ± 12.9 task return versus 131.6 ± 10.9 for one robot. With three robots, the system becomes less capacity-bound and all policies approach high on-time fulfilment. These tests measure whether the formulation remains useful beyond a single robot, rather than relying only on one curated scenario.

4.1 Interpretation and limitations

The attention weights provide a useful diagnostic rather than a separate supervision signal: express orders receive approximately 2.0 times their uniform attention share, and orders likely to miss their truck are picked less than half as often as those with ample slack. The gain over Greedy comes from express orders and 21% less travel per order.

There are important limits to the claim. The warehouse is a simulator, so its demand model and reward weights are assumptions rather than measurements from a live facility. The policy is trained on a finite family of layouts and does not solve arbitrary warehouse maps without retraining or transfer. Finally, the three-robot experiment uses a shared policy and an order-claiming mechanism; it is evidence of scalability, not a proof of optimal multi-agent coordination. These limitations motivate future work with logged warehouse traces, learned demand calibration, communication costs, and safety evaluation on physical robots.

4.2 Why reinforcement learning is appropriate

The best action depends on delayed consequences: picking an order now consumes capacity, changes the route, affects battery feasibility, and may protect or miss a future truck. The arrival process is only partially predictable because departure jitter is hidden. Consequently, a supervised target action is not available for each state, and a static shortest-path or priority rule cannot optimize the long-term trade-off. PPO learns this trade-off from repeated interaction using only rewards generated by the environment. The heuristic policies are therefore useful controls, while the proposed decision-maker remains an RL policy.

All code, configurations, trained checkpoints, evaluation data, figures, and complete reproduction commands are provided in the GitHub README.

References

1. Schulman et al., "Proximal Policy Optimization Algorithms," arXiv:1707.06347, 2017.
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